CS 5782 · DEEP LEARNING · CORNELL UNIVERSITY · SPRING 2026

TEAM MEMBERS

Rohan Modi

Ishaan Agarwal

Tree of *Thoughts*

Deliberate Problem Solving with Large Language Models · Yao et al., NeurIPS 2023

"Don't just chain it — tree it."

CornellBowers College of Computing + Information Science

INTRODUCTION & MOTIVATION

Why Do LLMs Struggle with Hard Problems?

- LLMs decode **token-by-token** , left-to-right, no lookahead or planning.
- Chain-of-Thought adds steps but remains **a single linear path.**
- Real problems require **exploration, backtracking, and evaluation.**
- Humans use **System 2 thinking**: slow and deliberate.

> **Goal:** Implement & analyze Tree of Thoughts, enabling LLMs to search over a tree of intermediate reasoning steps, not just a single chain.

BACKGROUND: PROMPTING PARADIGMS

From Chains to Trees

IO — Single input → output → **CoT** — Linear chain of steps → **ToT ✦** — Tree search over thoughts

Method	Explore	Backtrack	Plan
IO Prompting	✗	✗	✗
CoT	✗	✗	~
Tree of Thoughts ✦	✓	✓	✓

EVALUATION TASKS

Four Benchmarks

Math Dataset Difficult Math
Competition-level mathematical theorems.

GSM8K Mathematical Reasoning 8th Grade Math
Grade-school math word problems.

Mini Crosswords Constraint
5×5 crossword. DFS with letter-intersection tracking.

Game of 24 Puzzle
4 integers to equal 24 using basic arithmetic

METHODOLOGY

The ToT Framework

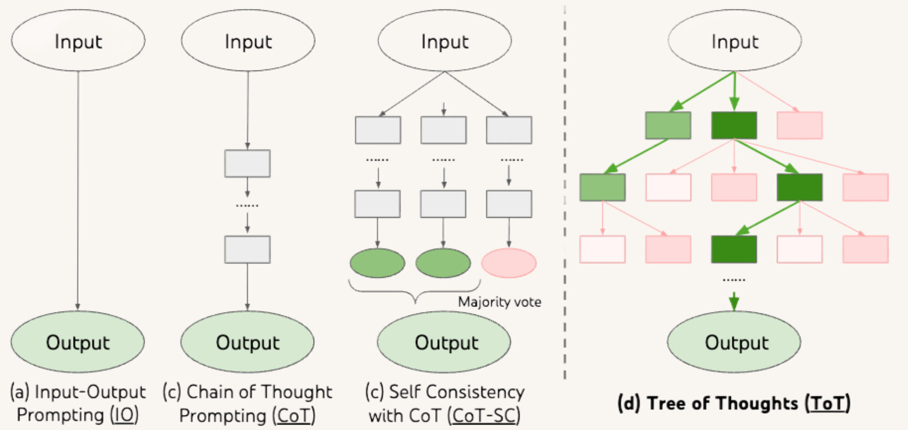

Fig. 1: *ToT maintains a tree of coherent "thoughts." Generator samples candidates; Evaluator scores each node; pruned branches (dashed) are abandoned. Search finds the optimal path.*

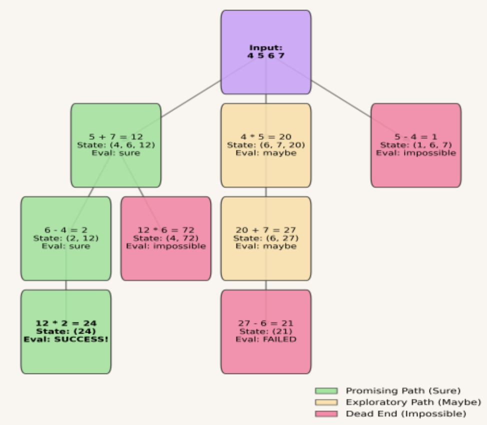

Fig. 2: *Example of a Game of 24 Tree*

FOUR FRAMEWORK COMPONENTS

01 Thought Decomposition
Break into meaningful intermediate steps at right granularity

02 Thought Generator
Sample k independent thoughts or propose sequentially

03 State Evaluator
LLM judges partial solutions via scoring or voting

04 Search Algorithm
BFS or DFS over the thought tree with pruning & backtracking

BFS
Best b candidates per step. Good for short-horizon tasks like simple math problems.

DFS
Depth-first with pruning. Efficient for deep, constrained spaces like crosswords.

OUR IMPLEMENTATION & EXTENSIONS

- Reimplemented ToT from scratch with **Gemini 2.5 – Flash.**
- Ablations on thought count k, beam width b, and evaluator prompt design
- **Cross-Domain Benchmarking:** Extended evaluation beyond the original paper's scope to include **GSM8K** and **MATH** to test the universal limits of search-based reasoning.

RESEARCH QUESTIONS

RQ1: When does branching search become necessary?
RQ2: What is the cost vs. performance trade-off of search?
RQ3: How does search intensity scale with difficulty?

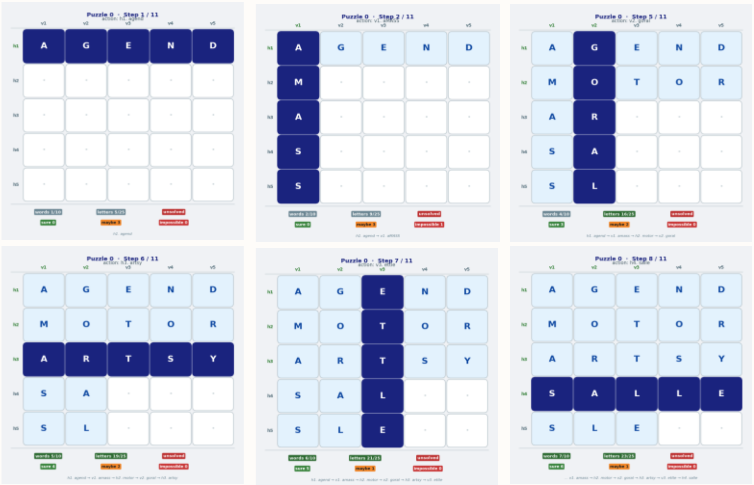

Fig. 3: *Example of a Crossword Tree*

RESULTS

Experimental Findings

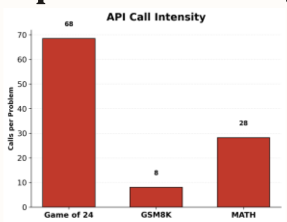

Fig. 4: *Computational intensity per problem*

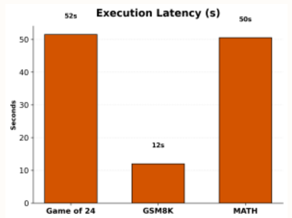

Fig. 5: *Average deliberation time*

Task	Method	Original Paper (ToT)	% Gemini 2.5 Flash
Crossword	Letter-level (ToT)	78.00%	69.6%
Crossword	Word-level (ToT)	60.00%	44%
Game of 24	Tree of Thoughts (ToT)	74.00%	46.7%
Game of 24	Chain of Thought (CoT)	4.00%	10%
GSM8K	Input-Output (IO)	-	85%
GSM8K	Chain of Thought (CoT)	-	90%
GSM8K	Tree of Thoughts (ToT)	-	100%
MATH	Input-Output (IO)	-	84%
MATH	Chain of Thought (CoT)	-	88%
MATH	Tree of Thoughts (ToT)	-	88%

Table 1: *Comparative summary of accuracy and compute across all prompting strategies.*

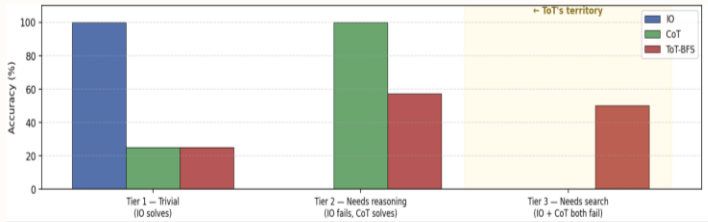

Fig. 6: *ToT's Territory: Tree of Thoughts outshines standard IO and CoT prompting in harder math problems*

CONCLUSION

Key Takeaways

- **ToT enables deliberate reasoning:** exploration + backtracking unavailable in CoT.
- Large gains on structured tasks; creative tasks benefit moderately
- Cost is the main trade-off: ToT requires significantly more LLM calls than CoT.

FUTURE WORK

- **Learned evaluators** trained on labeled reasoning traces.
- **MCTS integration:** Monte Carlo Tree Search for better exploration.

REFERENCES

[1] Yao et al. "Tree of Thoughts: Deliberate Problem Solving with LLMs." *NeurIPS 2023.*